

Constructing More Super-optimal Pairings via Small Degree Endomorphisms

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Abstract. A variety of cryptographic applications are enabled by the properties of bilinear pairings on elliptic curves. Thus, efficient pairing computation is of crucial importance for practical deployments of pairing-based cryptographic protocols. One of the most important technical approaches is to construct pairing variants with short Miller loops in efficiency. Driven by this motivation, the variants of Tate pairings such as the ate and the optimal ate pairings, are successively proposed. Moreover, on specific pairing-friendly curves admitting efficiently computable automorphisms, the number of basic Miller iterations can be further reduced to $\log_2(r)/2\varphi(k)$, where φ and k denote the Euler's totient function and the embedding degree with respect to r , respectively. Such pairings are named as the super-optimal ate pairings. Nevertheless, several curves are not compatible with the existing methods for constructing super-optimal ate pairings as they admit no non-trivial automorphisms. This paper aims to establish a framework for constructing super-optimal ate pairings on more pairing-friendly curves employing degree- n endomorphisms with $n > 1$. In practice, small degree endomorphisms (e.g. $n = 2, 3$) can be computed efficiently and are primarily considered for enhancing the performance of pairing computations. More precisely, pairing-friendly curves with CM-discriminants 7 and 11 are equipped with degrees-2 and 3 endomorphisms, respectively, possessing the potential for constructing super-optimal ate pairings. For example, we exploit degree-2 endomorphism (resp. degree-3 endomorphism) and provide explicit formulas for the super-optimal ate pairings on family GG22D7 with CM-discriminant $D = 7$ and embedding degree $k = 22$ (resp. GG28D11 with CM-discriminant $D = 11$ and embedding degree $k = 28$). Additionally, we select a pairing-friendly curve GG22D7-457 at the 192-bit security level for implementation, and present detailed computational procedure, concrete cost analysis, together with the experimental results. The results illustrate that our new super-optimal pairing formula derives a saving of about **31.9%** $\mathbb{F}_p$ -multiplications for the Miller loop on GG22D7-457 compared to the previous method. In terms of CPU

clock cycles, leveraging our new formula is **26.0%** faster. This work expands the application of super-optimal pairings, and thus offer a broader range of choices of curves for pairing-based cryptography.

Keywords: Pairing-friendly curves Optimal pairing Super-optimal ate pairing Efficiently computable endomorphisms

1 Introduction

In recent years, pairings become a significant component of public-key cryptography due to their applications in various protocols, such as identity-based encryption [7], short signature [8], key agreements [10,26,36] and SNARKs (Succinct Non-interactive ARguments of Knowledge) [4,14,24]. A pairing on an elliptic curve over a finite field $\mathbb{F}_p$ is an efficient nondegenerate bilinear map $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ (or $e : \mathbb{G}_2 \times \mathbb{G}_1 \rightarrow \mathbb{G}_T$), where the groups $\mathbb{G}_1$ and $\mathbb{G}_2$ consist of the points on E with prime order r . While $\mathbb{G}_T$ is the group of r -th roots of unity in $\mathbb{F}_{p^k}^*$, where k denotes the embedding degree with respect to r .

In pairing-based cryptographic systems, the Weil and Tate pairings serve as fundamental tools and have been deeply investigated. The Tate pairings exhibit better performances in most practical scenarios since they only require one evaluation of the Miller function. As a result, a significant body of research has focused on the optimization of the Tate pairing computation. One of the main research goals is to shorten the length of the Miller loop, which is a crucial component in the pairing computations. Duursma and Lee [13], along with Barreto *et al.* [5], have made notable improvements, shortening the length of the Miller loops for the Tate pairings on supersingular abelian varieties by leveraging the η_T method.

In 2006, Hess *et al.* [23] extended this approach to a broader class of ordinary curves. Using the Frobenius endomorphism, they proposed the ate pairing with a shorter Miller loop, which further expanded the scope of research for pairing-based cryptographic. Following this, several variants of the ate pairing [28,30,44] have been successively proposed with the aim of further minimizing the length of the Miller loop, thereby enhancing the efficiency of the pairing computation.

In 2008, Vercauteren [40] introduced the optimal ate pairing, which can be computed within $\log_2(r)/\varphi(k)$ basic Miller iterations, where φ represents the Euler's totient function. Furthermore, in specific situations the bit-length of Miller loop can be further shortened to $\log_2(r)/2\varphi(k)$ by utilizing efficiently computable automorphisms [18,19]. This optimized variant is referred to as the **super-optimal ate pairing** [22,39].

Hess [22] presented the following two situations that we can construct the super-optimal pairings:

1. k is odd and $\#\text{Aut}(E) = 4$ ($j(E) = 1728$),
2. $k \nmid 3$ and $\#\text{Aut}(E) = 6$ ($j(E) = 0$),

where $\text{Aut}(E)$ denotes the automorphism group of E . The author also pointed out that in all other cases, the length of Miller loop can only be reduced to $r^{\frac{1}{\varphi(k)}}$. Subsequently, several works have concentrated on constructing super-optimal pairings over specific pairing-friendly curves equipped with efficiently computable automorphisms.

In [12,17], the authors constructed the super-optimal pairing on curve BW13-310 at the 128-bit security level by utilizing the non-trivial automorphism, and provided efficient algorithms for implementation. Dai *et al.* [11] revisited the super-optimal pairings on pairing-friendly curves with embedding degrees $k = 10, 14$ and made an efficient implementation. Up to now, the super-optimal pairings in the literature are constructed on curves with CM-discriminant $D = 1$ or 3 via automorphisms.

Nevertheless, not all pairing-friendly curves admit non-trivial automorphisms, such as those with $D \neq 1, 3$. A natural question is that whether we can construct super-optimal pairings on these curves. To the best of our knowledge, efficiently computable degree- n endomorphisms (GLV-endomorphisms [19]) have been leveraged to speed up the computations of Tate pairings on Cocks-Pinch curves with small embedding degrees [25]. Motivated by this work, we investigate to establish a framework for constructing super-optimal ate pairings on more generic families of pairing-friendly curves equipped with degree- n endomorphisms. Two specific situations $n = 2$ and 3 are detailedly discussed. The pairing-friendly curves with CM-discriminant $D = 2$ or 7 (resp. $D = 11$) possess potential for the constructions of super-optimal ate pairings utilizing degree-2 endomorphisms (resp. degree-3 endomorphisms). Precisely, we implement our improved super-optimal ate pairing on curve GG22D7-457 admitting a degree-2 endomorphism at the 192-bit security level proposed by Gasnier and Guillevic [20]. Concrete computational cost analysis is also provided. The experimental results show that our new formula for the super-optimal pairing outperforms the optimal ate pairing on GG22D7-457.

1.1 Contributions

The contributions of this paper are summarized as follows:

1. We determine the pairing-friendly curves which are equipped with degrees- n endomorphisms, and provide concrete formulas for the super-optimal pairings. Besides, we exploit the above method to the GLV-curves admitting degrees-2, 3 endomorphisms and obtain the corresponding super-optimal pairings. Hence, our techniques can be potentially exploited to extend the applications of super-optimal pairings in elliptic curve cryptography.
2. We employ degrees-2 and 3 endomorphisms to accelerate the pairing computations on curves with CM-discriminants $D = 7$ and 11, respectively. Particularly, we present explicit formulas for the super-optimal pairings on families GG22D7 and GG28D11. Furthermore, we provide concrete implementation details and cost analysis on curve GG22D7-457. Our source code is available at

<https://github.com/zyuh6/relic-GG22D7>.

For the computation of Miller loop, our technique demonstrates a saving of about **31.9%** in terms of $\mathbb{F}_p$ -multiplications compared to the optimal pairing. The experimental results illustrate that our new formula for super-optimal pairing is **26.0%** faster than the previous approach in terms of CPU clock cycles.

1.2 Comparison with the previous literature

The first novel utilization of degree- n endomorphisms with $n > 1$ for pairing computation is due to Ionica in [25]. The authors leveraged efficiently computable endomorphisms to accelerate the computations of Tate pairings with input subgroup $\mathbb{G}_1 \times \mathbb{G}_2$, and shortened the length of Miller loop to around $r^{\frac{1}{2}}$. Furthermore, it only focuses on the Cocks-Pinch curves with small embedding degrees k , especially for $k = 2$.

In this paper we concentrate on establishing a framework for the constructions of super-optimal ate pairings with input subgroup $\mathbb{G}_2 \times \mathbb{G}_1$ employing degree- n endomorphisms. To this end, we make use of both degree- n endomorphisms and Frobenius map to further reduce the length of Miller loop to $r^{\frac{1}{2\varphi(k)}}$. This optimization expands the applications for super-optimal pairings on a broader class of pairing-friendly curves without non-trivial automorphisms, and has indeed benefited the pairing computation on actual families with CM-discriminants 7 and 11. The scope of this paper is thus orthogonal to the previous literature.

1.3 Organizations of this paper

The notations and definitions are stated in Section 2. The Miller algorithm and the optimal ate pairing are recalled. Our formulas for super-optimal pairings, along with a explicit theoretical analysis are presented in Section 3. Section 4 illustrates the concrete implementation details, computational cost analysis and experimental results. Finally, our conclusion and future work are drawn in Section 5.

2 Preliminaries

In this section, we introduce the corresponding mathematical preliminaries required in this paper. Let E denote a short Weierstrass curve over a finite field $\mathbb{F}_p$, where $p > 5$ is a prime. The additive group $E(\mathbb{F}_p)$ consists of points (x, y) satisfying the following equation

$$E/\mathbb{F}_p : y^2 = x^3 + a_4x + a_6$$

and a point at infinity $\mathcal{O}_E$. The j -invariant of E is defined as $j(E) = 1728 \cdot \frac{4a_4^3}{4a_4^3 + 27a_6^2}$. Denote by $\#E(\mathbb{F}_p)$ the cardinality of $E(\mathbb{F}_p)$. It satisfies that $\#E(\mathbb{F}_p) =$

$p + 1 - t$ [41, Theorem 4.12], where t is the trace of the p -power Frobenius endomorphism $\pi : (x, y) \mapsto (x^p, y^p)$.

Let r be a large prime factor of $\#E(\mathbb{F}_p)$. The embedding degree k with respect to r is the smallest positive integer such that $r \mid p^k - 1$. We denote the eigenspaces of π acting on the r -torsion group $E[r] = \{P \in E \mid [r]P = \mathcal{O}_E\}$ with eigenvalues 1 and p by $\mathbb{G}_1$ and $\mathbb{G}_2$, respectively. In other words, $\mathbb{G}_1$ and $\mathbb{G}_2$ can be represented as

$$\mathbb{G}_1 = E(\mathbb{F}_p)[r], \quad \mathbb{G}_2 = E(\mathbb{F}_{p^k})[r] \cap \{P \in E \mid \pi(P) = [p]P\}.$$

In the following, we introduce the concepts of endomorphisms of ordinary elliptic curves, the families of pairing-friendly curves and the optimal ate pairing.

2.1 Endomorphism rings of ordinary curves

With the same notation as above, and let D be a positive square-free integer such that $4p - t^2 = Dy^2$, $y \in \mathbb{Z}$. From [41, Theorem 10.6], we know that the endomorphism ring of an ordinary curve E denoted by $\text{End}(E)$ is isomorphic to an order of an imaginary quadratic field

$$K = \mathbb{Q}(\sqrt{-D}) = \{a + b\sqrt{-D} \mid a, b \in \mathbb{Q}\}.$$

The maximal order of $\mathbb{Q}(\sqrt{-D})$ is a finitely generated abelian group [41]

$$\mathcal{O}_K = \begin{cases} \mathbb{Z} \left[\frac{1+\sqrt{-D}}{2} \right], & D \equiv 3 \pmod{4}, \\ \mathbb{Z} [\sqrt{-D}], & D \equiv 1, 2 \pmod{4}. \end{cases}$$

An **order** of K is a subring R such that $\mathbb{Z} \subsetneq R \subseteq \mathcal{O}_K$ [41]. Such an order has the form

$$R = f\mathcal{O}_K,$$

where the integer f is called the conductor of R . Let $\sigma \in \text{End}(E)$ denote an endomorphism of E . If σ is simultaneously an degree- n endomorphism, then it satisfies the following equation

$$N(\sigma) = \sigma\hat{\sigma} = n, \tag{1}$$

where $N(\sigma)$ and $\hat{\sigma}$ represent the norm and conjugate element of σ , respectively.

If $D \equiv 1, 2 \pmod{4}$, then σ has the form $\sigma = a + b\sqrt{-D}$, $a, b \in \mathbb{Z}$, and Eq. (1) becomes

$$\sigma\hat{\sigma} = (a + b\sqrt{-D})(a - b\sqrt{-D}) = a^2 + b^2D = n.$$

If $D \equiv 3 \pmod{4}$, then σ can be expressed as $\sigma = \frac{a}{2} + \frac{b\sqrt{-D}}{2}$, $a, b \in \mathbb{Z}$, then Eq. (1) is

$$\sigma\hat{\sigma} = \left(\frac{a}{2} + \frac{b\sqrt{-D}}{2}\right)\left(\frac{a}{2} - \frac{b\sqrt{-D}}{2}\right) = \frac{a^2}{4} + \frac{b^2D}{4} = n.$$

A curve with CM-discriminant D satisfying Eq. (1) admits a degree- n endomorphism σ . Once we derive the integer solution of this equation, we can determine the form of σ . For simplicity, we represent the degree- n endomorphism as $\sigma = a + b\sqrt{-D}$, where $a, b \in \frac{1}{2}\mathbb{Z}$.

2.2 Families of pairing-friendly elliptic curve

In this subsection, we present the definitions of the CM discriminant D and the families of pairing-friendly curves.

An elliptic curve E over $\mathbb{F}_p$ is pairing-friendly if the following two conditions hold:

1. the order of the additive group $\#E(\mathbb{F}_p)$ has a large prime factor r satisfying $\frac{\log_2(p)}{\log_2(r)} \leq 2$,
2. the embedding degree k with respect to r is less than $\log_2(r)/8$.

The tuple (p, t, r) of a pairing-friendly curve can be parametrized by polynomials. We state the following definition for illustration.

Definition 1. *Let $p(x), t(x), r(x) \in \mathbb{Q}[x]$ be non-zero polynomials. A polynomial tuple $(p(x), t(x), r(x))$ parametrizes a family of pairing-friendly elliptic curves with embedding degree k and CM discriminant D , if the following conditions are satisfied:*

1. $p(x)$ represents primes and it is non-constant, irreducible, with a positive leading coefficient.
2. $r(x)$ is a non-constant, irreducible, integer-valued polynomial with a positive leading coefficient.
3. $r(x)$ divides both $p(x) + 1 - t(x)$ and $\Phi_k(t(x) - 1)$, where $\Phi_k(x)$ denotes the k -th cyclotomic polynomial.
4. There are infinitely many integer solutions (x, Y) for the parametrized CM equation $DY^2 = 4p(x) - t(x)^2$.

In pairing-based cryptography, appropriately increasing the embedding degree k allows for a smaller characteristic p , resulting in more efficient pairing computations. Hence, we consider pairing-friendly curves with embedding degrees much greater than 2 in this paper.

2.3 Optimal ate pairing and super-optimal ate pairing

We first present the definitions of the Miller function $f_{n,P}$ and Miller's algorithm. For any point $P \in E$ and $n \in \mathbb{Z}$, denote by $f_{n,P}$ the normalized rational function associated with the divisor

$$(f_{n,P}) = n(P) - ([n]P) - (n-1)(\mathcal{O}_E).$$

Especially for $P \in E[r]$, the corresponding divisor becomes

$$(f_{r,P}) = r(P) - r(\mathcal{O}_E).$$

For any $i, j \in \mathbb{Z}$, there exists a relationship between $f_{i,P}, f_{j,P}$ and $f_{i+j,P}$

$$(f_{i+j,P}) = \left(f_{i,P} \cdot f_{j,P} \cdot \frac{l_{[i]P, [j]P}}{\nu_{[i+j]P}} \right), \quad (2)$$

where $l_{[i]P, [j]P}$ denotes a straight line passing through the points $[i]P$ and $[j]P$, and $\nu_{[i+j]P}$ represents a vertical line passing through $[i+j]P$ and $[-i-j]P$. A well-known efficient algorithm for evaluating $f_{n,P}(Q)$, where $n \in \mathbb{Z}$, $P \in \mathbb{G}_1$ and $Q \in \mathbb{G}_2$ was proposed by Miller in [31].

Let $\lambda = \ell r$ with $r \nmid \ell$, and write the p -adic representation of λ as $\lambda = \sum_{i=0}^{\varphi(k)-1} c_i p^i$. By Minkowski's theorem [32], there exists one of the short vectors $V = (c_0, \dots, c_{\varphi(k)-1})$ with $|c_i| \leq r^{1/\varphi(k)}$ such that $\lambda \equiv 0 \pmod{r}$, where $\varphi(k)$ is the Euler's totient function. Define μ_r to be the group of r -th roots of unity. An optimal ate pairing [40] $opt : \mathbb{G}_2 \times \mathbb{G}_1 \rightarrow \mu_r$ on E is defined as follows:

$$(Q, P) \mapsto \left(\prod_{i=0}^{\varphi(k)-1} f_{c_i, Q}^{p^i}(P) \cdot \prod_{i=0}^{\varphi(k)-2} \frac{l_{[s_{i+1}]Q, [c_i p^i]Q}(P)}{\nu_{[s_i]Q}(P)} \right)^{\frac{p^k-1}{r}} \quad (3)$$

with $s_i = \sum_{j=i}^{\varphi(k)-1} c_j p^j$. The bilinearity of opt can be established by leveraging the bilinearity of ate pairing [23] and Eq. (2). This pairing is nondegenerate if $\ell k p^{k-1} \not\equiv ((p^k - 1)/r) \cdot \sum_{i=0}^{\varphi(k)-1} i c_i p^{i-1} \pmod{r}$.

In specific situations, it is feasible to incorporate efficiently computable automorphisms to further shorten the lengths of Miller loops. Pairings for which the length of Miller loop is shorter than $r^{1/\varphi(k)}$ are called **super-optimal (ate) pairings** [17, 22, 39].

2.4 The utilization of endomorphisms to accelerate the computations of Tate pairings

In this subsection, we introduce the utilization of degree- n endomorphisms for the improvement of Tate pairings [25]. With the same notations as Section 2.1, and let σ denote a degree- n endomorphism with respect to $a + b\sqrt{-D}$ in $\text{End}(E)$. Then the characteristic equation of σ is given by

$$\sigma^2 + tr(\sigma)\sigma + N(\sigma) = \sigma^2 - 2a\sigma + n = 0, \quad (4)$$

where $tr(\sigma) = \sigma + \hat{\sigma}$ is the trace of σ . Assume that $P \in \mathbb{G}_1, Q \in \mathbb{G}_2$. It follows from [25, Theorem 2] that P and Q are two eigenvectors of σ . Therefore, there exists an integer λ such that $\sigma(P) = [\lambda]P$. Following the notation established in [25], we define

$$corr_{P_1, P_2} := \frac{l_{P_1, P_2}}{\nu_{P_1 + P_2}}$$

with divisor $(P_1) + (P_2) - (P_1 + P_2) - (\mathcal{O}_E)$ to be the correction of two points P_1 and P_2 throughout the remainder of this paper. It can be deduced that if P_1 or P_2 is a point of infinity, then $corr_{P_1, P_2}$ corresponds to a zero divisor and thus represents a constant which can be normalized to 1. The following lemma characterizes the relation between the Miller functions $f_{\lambda, P}$ and $f_{\lambda, \sigma(P)}$.

Lemma 1. [25, Lemma 1] *Let E be an elliptic curve over a finite field $\mathbb{F}_q$ and σ an endomorphism of E with degree n . Let $P \in \mathbb{G}_1$, $Q \in \mathbb{G}_2$. Then for any integer λ the following equality holds up to a constant:*

$$f_{\lambda, \sigma(P)}(\sigma(Q)) = f_{\lambda, P}^n(Q) \cdot \prod_{T \in \ker(\sigma) \setminus \{\mathcal{O}_E\}} \frac{\text{corr}_{[\lambda]P, T}(Q)}{\text{corr}_{P, T}^\lambda(Q)},$$

where $\ker(\sigma)$ is the kernel of σ with $\#\ker(\sigma) = \deg(\sigma) = n$.

Proof. By the definition of pullback action, we have

$$\begin{aligned} & \sigma^*(f_{\lambda, \sigma(P)}) \\ &= \lambda \sum_{T \in \ker(\sigma)} (P + T) - \sum_{K \in \ker(\sigma)} ([\lambda]P + T) - (\lambda - 1) \sum_{K \in \ker(\sigma)} (T) \\ &= \lambda \sum_{T \in \ker(\sigma)} ((P + T) - (T)) - \sum_{T \in \ker(\sigma)} (([\lambda]P + T) - (T)) \\ &= \sum_{T \in \ker(\sigma)} (\lambda(P) - \lambda(\mathcal{O}_E) - (\text{corr}_{P, T}^\lambda)) - \sum_{T \in \ker(\sigma)} (([\lambda]P) - (\mathcal{O}_E) - (\text{corr}_{[\lambda]P, T})) \\ &= \sum_{T \in \ker(\sigma)} (\lambda(P) - ([\lambda]P) - (\lambda - 1)(\mathcal{O}_E)) + \sum_{T \in \ker(\sigma)} ((\text{corr}_{[\lambda]P, T}) - (\text{corr}_{P, T}^\lambda)). \end{aligned}$$

Recall that $\#\ker(\sigma) = n$ and $\text{div}(\text{corr}_{P, \mathcal{O}_E}) = \text{div}(\text{corr}_{[\lambda]P, \mathcal{O}_E}) = 0$, it yields that

$$\sigma^*(f_{\lambda, \sigma(P)}) = \text{div} \left(f_{\lambda, P}^n \cdot \prod_{T \in \ker(\sigma) \setminus \{\mathcal{O}_E\}} \frac{\text{corr}_{[\lambda]P, T}}{\text{corr}_{P, T}^\lambda} \right) = (f_{\lambda, \sigma(P)} \circ \sigma),$$

which completes the proof. $\square$

By Lemma 1 and the characteristic equation of σ , one can derive a new algorithm for Tate pairing computation, whose length of Miller loop is halved. See [25, Theorem 3] for more details. As the degree n becomes larger, the computational complexity for the correction accordingly increases. Therefore, in practical applications, the endomorphisms with relatively low degrees (e.g. $n = 2, 3$) are mainly considered.

3 Main Results

In this section, we propose improved formulas for super-optimal ate pairings on a broader class of pairing-friendly curves. We establish the existence conditions for super-optimal ate pairings $\text{sopt}: \mathbb{G}_2 \times \mathbb{G}_1 \rightarrow \mu_r$, and derive the corresponding formulas leveraging degree- n endomorphisms. In addition, the formulas are efficient when n is relatively small. Consequently, we provide explicit formulas for super-optimal pairings on two cryptographically interesting families GG22D7 and GG28D11 [20] by employing degrees-2 and 3 endomorphisms, respectively.

3.1 Constructing the super-optimal ate pairings on $\mathbb{G}_2 \times \mathbb{G}_1$ by utilizing degree- n endomorphisms

In this subsection, we present the formulas for super-optimal pairings on GLV-curves admitting degree- n endomorphisms σ . Let the same notations be as above. Define $\sigma = a + b\sqrt{-D}$ to be a degree- n endomorphism satisfying Eq. (4). If $n = 1$, σ naturally denotes an efficiently computable automorphism, which has been widely used to construct super-optimal pairings on curves with $D = 1$ or 3 [11, 17]. In the following, we only discuss the endomorphisms with degrees $n > 1$.

By [25, Theorem 2] one can prove that there exists an integer s such that $\sigma(Q) = [s]Q$ for $Q \in \mathbb{G}_2$. Consider the composite map $\tau = \sigma \circ \pi$ associated to $\alpha = (a + b\sqrt{-D}) \cdot \zeta_k \in \mathbb{Q}(\zeta_k, \sqrt{-D})$, where ζ_k denotes the primitive k -th root of unity. Then we deduced that

$$\tau(Q) = \sigma \circ \pi(Q) = [s \cdot p \bmod r]Q = [\lambda]Q, \quad Q \in \mathbb{G}_2, \quad \lambda \in \mathbb{Z}.$$

The following theorem characterizes the conditions for the existence of super-optimal ate pairings, and presents a framework for their constructions.

Theorem 1. *Let the notations be as above. Assume that E is an ordinary pairing-friendly curve possessing CM-discriminant D , embedding degree k with respect to r , and a small degree- n GLV-endomorphism $\sigma = a + b\sqrt{-D}$ ($n > 1$). Denote by $\hat{\sigma}$ the conjugate element of σ . Let $P \in \mathbb{G}_1$ and $Q \in \mathbb{G}_2$. If it satisfies that:*

1. $\sqrt{-D} \notin \mathbb{Q}(\zeta_k)$, together with $a \neq 0$ or $k \not\equiv 0 \pmod{4}$,
2. $\tau(Q) = \sigma \circ \pi(Q) = [\lambda]Q$, where $\lambda \in \mathbb{Z}$ and $\lambda = \mathcal{O}\left(r^{\frac{1}{2\varphi(k)}}\right)$,

then there exist a polynomial $h(x) = \sum_{i=0}^{2\varphi(k)-1} h_i x^i$ with $h_i = \mathcal{O}\left(r^{\frac{1}{2\varphi(k)}}\right)$ such that $h(\lambda) = \ell r$, $\ell \in \mathbb{Z}$, and a super-optimal ate pairing $\text{sopt} : \mathbb{G}_2 \times \mathbb{G}_1 \rightarrow \mu_r$

$$\left(f_{\lambda, Q}^{m_1}([n]P) \cdot \prod_{i=1}^{2\varphi(k)-1} f_{h'_i, Q}^{n p^i}(\hat{\sigma}(P)) \cdot \text{CORR}_{4,i}^{p^i}(P) \cdot \prod_{i=0, i \neq 1}^{2\varphi(k)-1} f_{h''_i, Q}^{p^i}([n]P) \cdot \prod_{i=1}^3 \text{CORR}_i([n]P) \right)^{\frac{p^k-1}{r}},$$

where $m_0 = \sum_{i=0, i \neq 1}^{2\varphi(k)-1} h''_i p^i$, $m_1 = \sum_{i=1}^{2\varphi(k)-1} h'_i p^{i-1}$, $h''_0 = h_0$, $h'_1 = h_1$, $h'_i = h_i C_i(a, n)$, $h''_i = h_i D_i(a, n)$ with $i \geq 2$, and C_i, D_i are polynomials in a, n . The corrections are defined as follows:

$$\begin{aligned} \text{CORR}_1(P) &= \nu_{[m_0]Q}(P), \quad \text{CORR}_2(P) = \prod_{i=1}^{2\varphi(k)-2} \text{corr}_{[\sum_{j=i+1}^{2\varphi(k)-1} h'_j p^{j-1}]Q, [h'_i p^{i-1}]Q}(P), \\ \text{CORR}_3(P) &= \prod_{i=0, i \neq 1}^{2\varphi(k)-2} \text{corr}_{[\sum_{j=i+1}^{2\varphi(k)-1} h''_j p^j]Q, [h''_i p^i]Q}(P), \\ \text{CORR}_{4,i}(P) &= \prod_{T \in \ker(\sigma) \setminus \{\mathcal{O}_E\}} \frac{\text{corr}_{[h'_i]Q, T}(\hat{\sigma}(P))}{\text{corr}_{h'_i, T}(\hat{\sigma}(P))}. \end{aligned}$$

Furthermore, this pairing is nondegenerate if

$$\ell k p^{k-1} \not\equiv ((p^k - 1)/r) \cdot \left(\lambda \sum_{i=1}^{2\varphi(k)-1} i h'_i p^{i-2} + \sum_{i=0, i \neq 1}^{2\varphi(k)-1} i h''_i p^{i-1} \right) \pmod{r},$$

where ℓ is an integer such that $m_0 + m_1 \lambda = \ell r$.

Before proving this theorem, we require the following proposition.

Proposition 1. *With the notation as above, and let $Q \in \mathbb{G}_2$. Assume that $\sigma \circ \pi(Q) = [\lambda]Q$, then we have*

$$\lambda^2 - 2ap\lambda + np^2 \equiv 0 \pmod{r}.$$

Proof. Based on the above analysis, $\tau = \sigma \circ \pi$ corresponds to the element $\alpha = (a + b\sqrt{-D}) \cdot \zeta_k \in K$. By Eq. (4) we can derive the following relation

$$\tau^2 - 2ap\tau + np^2 = \zeta_k^2(\sigma^2 - 2a\sigma + n) = 0.$$

Since $\tau(Q) = [\lambda]Q$, it holds that

$$[\lambda^2 - 2ap\lambda + np^2]Q = \mathcal{O}_E.$$

From $Q \in \mathbb{G}_2$ we deduce

$$\lambda^2 - 2ap\lambda + np^2 \equiv 0 \pmod{r}.$$

This ends the proof. □

We are now in a position to provide the whole proof of Theorem 1.

Proof of Theorem 1 According to [42, Theorem 2.4], if it satisfies that

$$\sqrt{-D} \notin \mathbb{Q}(\zeta_k), \text{ along with } a \neq 0 \text{ or } k \not\equiv 0 \pmod{4},$$

then $\alpha = (a + b\sqrt{-D}) \cdot \zeta_k$ is the primitive root of $\mathbb{Q}(\zeta_k, \sqrt{-D})$. This implies that

$$\deg(f(x)) = [\mathbb{Q}(\alpha) : \mathbb{Q}] = [\mathbb{Q}(\zeta_k, \sqrt{-D}) : \mathbb{Q}] = 2\varphi(k),$$

where $f(x)$ denotes the minimal polynomial of α (with respect to τ). From $\tau(Q) = [\lambda]Q$ we have $[f(\lambda)]Q = \mathcal{O}_E$ for $Q \in \mathbb{G}_2$, yielding $f(\lambda) \equiv 0 \pmod{r}$.

On this basis, by Minkowski's theorem [32] and LLL's algorithm [29] there exists a dimension- $2\varphi(k)$ short vector $(h_0, \dots, h_{2\varphi(k)-1})$ with $h_i = \mathcal{O}\left(r^{\frac{1}{2\varphi(k)}}\right)$ such that

$$h(\lambda) = \sum_{i=0}^{2\varphi(k)-1} h_i \lambda^i \equiv 0 \pmod{r}.$$

Let $g(\lambda) = \lambda^2 - 2ap\lambda + np^2$. Then it follows from Proposition 1 that

$$m(\lambda) \equiv h(\lambda) \pmod{g(\lambda)} \equiv m_0 + m_1 \lambda \equiv 0 \pmod{r}. \quad (5)$$

We now determine the form of $m(\lambda)$. By $\lambda^2 \equiv 2ap\lambda - np^2 \pmod{g(\lambda)}$, every λ^i ($i \geq 1$) can be reduced to

$$\lambda^i \equiv c_i \lambda + d_i \pmod{g(\lambda)}.$$

Moreover, we can drive the following recursive relation

$$\begin{cases} c_1 = 1, d_1 = 0, \\ c_i = 2apc_{i-1} + d_{i-1}, d_i = -np^2c_{i-1}, i \geq 2. \end{cases}$$

From this relation we deduce that c_i and d_i can be re-expressed by

$$c_i = p^{i-1}C_i(a, n), d_i = p^iD_i(a, n),$$

where both C_i and D_i are polynomials in a and n .

Utilizing Eq. (5) yields

$$m_0 = \sum_{i=0, i \neq 1}^{2\varphi(k)-1} h_i'' p^i, m_1 = \sum_{i=1}^{2\varphi(k)-1} h_i' p^{i-1}, \quad (6)$$

where $h_0'' = h_0$, $h_1' = h_1$, and $h_i' = h_i C_i(a, n)$, $h_i'' = h_i D_i(a, n)$ for $i \geq 2$. Since a, n and k are extremely small compared to h_i , $C_i(a, n)$ and $D_i(a, n)$ can be regarded as $\mathcal{O}(1)$. Thus we obtain $h_i' = \mathcal{O}\left(r^{\frac{1}{2\varphi(k)}}\right)$, $h_i'' = \mathcal{O}\left(r^{\frac{1}{2\varphi(k)}}\right)$.

Let $t_r(Q, P) = f_{r, Q}(P)^{\frac{p^k-1}{r}}$ and $a_p(Q, P) = f_{p, Q}(P)^{\frac{p^k-1}{r}}$ represent the reduced Tate and ate pairings [23], respectively. Combining Eqs. (2) and (6), and using [40, Eqs. (4) and (5)] we deduce that

$$\begin{aligned} & t_r(Q, P)^{n\ell} \\ &= f_{m_0+m_1\lambda, Q}([n]P)^{\frac{p^k-1}{r}} \\ &= (f_{m_0, Q}([n]P) \cdot f_{m_1\lambda, Q}([n]P) \cdot \nu_{[m_0]Q}([n]P))^{\frac{p^k-1}{r}} \\ &= a_p(Q, P)^M \left(f_{\lambda, Q}^{m_1}([n]P) \cdot \prod_{i=1}^{2\varphi(k)-1} f_{h_i', [\lambda]Q}^{p^{i-1}}([n]P) \cdot \prod_{i=0, i \neq 1}^{2\varphi(k)-1} f_{h_i'', Q}^{p^i}([n]P) \cdot \prod_{i=1}^3 \text{CORR}_i([n]P) \right)^{\frac{p^k-1}{r}}, \end{aligned}$$

where

$$\begin{aligned} \text{CORR}_1(P) &= \nu_{[m_0]Q}(P), \text{CORR}_2(P) = \prod_{i=1}^{2\varphi(k)-2} \text{corr}_{[\sum_{j=i+1}^{2\varphi(k)-1} h_j' p^{j-1}]Q, [h_i' p^{i-1}]Q}(P), \\ \text{CORR}_3(P) &= \prod_{i=0, i \neq 1}^{2\varphi(k)-2} \text{corr}_{[\sum_{j=i+1}^{2\varphi(k)-1} h_j'' p^j]Q, [h_i'' p^i]Q}(P), \\ M &= n \left(\lambda \sum_{i=1}^{2\varphi(k)-1} i h_i' p^{i-2} + \sum_{i=0, i \neq 1}^{2\varphi(k)-1} i h_i'' p^{i-1} \right). \end{aligned}$$

Due to the bilinearities of $t_r(Q, P)$ and $a_p(Q, P)$, the following function

$$\begin{aligned} & \text{sopt}(Q, P) \\ &= \left(f_{\lambda, Q}^{m_1}([n]P) \cdot \prod_{i=1}^{2\varphi(k)-1} f_{h'_i, [\lambda]Q}^{p^{i-1}}([n]P) \cdot \prod_{i=0, i \neq 1}^{2\varphi(k)-1} f_{h''_i, Q}^{p^i}([n]P) \cdot \prod_{i=1}^3 \text{CORR}_i([n]P) \right)^{\frac{p^k-1}{r}} \end{aligned} \quad (7)$$

also gives a bilinear pairing. By exploiting Lemma 1, one can show that

$$\prod_{i=1}^{2\varphi(k)-1} f_{h'_i, [\lambda]Q}^{p^{i-1}}([n]P) = \prod_{i=1}^{2\varphi(k)-1} f_{h'_i, \sigma(Q)}^{p^i}(\sigma \circ \hat{\sigma}(P)) = \prod_{i=1}^{2\varphi(k)-1} \left(f_{h'_i, Q}^n(\hat{\sigma}(P)) \cdot \text{CORR}_{4,i}(P) \right)^{p^i},$$

$$\text{where } \text{CORR}_{4,i}(P) = \prod_{T \in \ker(\sigma) \setminus \{O_E\}} \frac{\text{corr}_{[h'_i]Q, T}(\hat{\sigma}(P))}{\text{corr}_{Q, T}^{h'_i}(\hat{\sigma}(P))}.$$

Plugging this equality into Eq. (7), we can obtain the final formula for $\text{sopt}(Q, P)$. Since $\lambda, h'_i, h''_i = \mathcal{O}\left(r^{\frac{1}{2\varphi(k)}}\right)$, this pairing is super-optimal. Additionally, $\text{sopt}(Q, P)$ degenerates when $t_r(Q, P)^{n\ell} = a_p(Q, P)^M$. It follows from [23, Theorem 1] and [40, Theorem 4] that

$$t_r(Q, P)^{n\ell} = a_p(Q, P)^{n\ell k p^{k-1}((p^k-1)/r)^{-1} \bmod r}.$$

Therefore, this super-optimal pairing is nondegenerate when

$$\ell k p^{k-1} \not\equiv ((p^k-1)/r) \cdot \left(\lambda \sum_{i=1}^{2\varphi(k)-1} i h'_i p^{i-2} + \sum_{i=0, i \neq 1}^{2\varphi(k)-1} i h''_i p^{i-1} \right) \bmod r,$$

which completes the proof. $\square$

The above theorem gives a framework to determine the existence of super-optimal pairings on ordinary curves admitting degree- n endomorphisms, and construct explicit formulas. In specific situations, we can further simplify the formulas in Theorem 1. Consider a family $(p(x), t(x), r(x))$ of pairing-friendly curves with embedding degree k , CM-discriminant D and a degree- n endomorphism $\sigma = a + b\sqrt{-D}$, constructed by KSS method [20, 27] (i.e. by taking $r(x) \in \mathbb{Q}[x]$ as the minimal polynomial of $\alpha = (a + b\sqrt{-D}) \cdot \zeta_k \in \mathbb{Q}(\zeta_k, \sqrt{-D})$). Denote this particular KSS family by

$$F(n, D, k, a, b).$$

Since $\sigma \circ \pi \sim \alpha \bmod r(x)$, we deduce that $\sigma \circ \pi(Q) = [x \bmod r(x)]Q$, $Q \in \mathbb{G}_2$. The following proposition gives corresponding formulas for the super-optimal pairing on family $F(n, D, k, a, b)$.

Proposition 2. *Let the notations be denoted as above. Assume that $\sqrt{-D} \notin \mathbb{Q}(\zeta_k)$, together with $a \neq 0$ or $k \not\equiv 0 \bmod 4$. Let $P \in \mathbb{G}_1$ and $Q \in \mathbb{G}_2$. Define $p(x), t(x)$ and $r(x)$ to be the parametrized polynomials of family $F(n, D, k, a, b)$. Denote the conjugate element of σ by $\hat{\sigma}$. Then the formula of the super-optimal*

ate pairing $sopt : \mathbb{G}_2 \times \mathbb{G}_1 \rightarrow \mu_r$ on the GLV-curves in $F(n, D, k, a, b)$ is presented as follows:

$$sopt(Q, P) = \left(f_{x,Q}^x([n]P) \cdot f_{x,Q}^{np}(\hat{\sigma}(P)) \cdot f_{-2ax,Q}^p([n]P) \cdot f_{n,Q}^{p^2}([n]P) \cdot CORR(P) \right)^{\frac{p^k-1}{r}},$$

where

$$CORR(P) = l_{[x^2]Q, [np^2]Q}([n]P) \cdot \left(\prod_{T \in \ker(\sigma) \setminus \{\mathcal{O}_E\}} \frac{corr_{[x]Q, T}(\hat{\sigma}(P))}{corr_{Q, T}^x(\hat{\sigma}(P))} \right)^p.$$

Let ℓ be an integer such that $x^2 - 2apx + np^2 = \ell r$. Moreover, if

$$\ell k p^{k-1} \not\equiv ((p^k - 1)/r) \cdot (np - 2ax) \pmod{r},$$

then the super-optimal ate pairing is nondegenerate.

Proof. Since $\sigma \circ \pi(Q) = [x]Q$, by Proposition 1 we have

$$x^2 - 2apx + np^2 \equiv 0 \pmod{r}.$$

Thus it holds that

$$\begin{aligned} & t_r(Q, P)^{n\ell} \\ &= f_{x^2-2apx+np^2, Q}([n]P)^{\frac{p^k-1}{r}} \\ &= (f_{x^2, Q}([n]P) f_{-2apx, Q}([n]P) f_{np^2, Q}([n]P) l_{[x^2]Q, [np^2]Q}([n]P))^{\frac{p^k-1}{r}} \\ &= \left(f_{x, Q}^x([n]P) f_{x, \sigma(Q)}^p([n]P) f_{-2ax, Q}^p([n]P) f_{n, Q}^{p^2}([n]P) \cdot L \right)^{\frac{p^k-1}{r}} \cdot a_p(Q, P)^{n^2 p - 2anx}, \end{aligned}$$

where $t_r(Q, P) = f_{r, Q}(P)^{\frac{p^k-1}{r}}$ and $a_p(Q, P) = f_{p, Q}(P)^{\frac{p^k-1}{r}}$ represent the reduced Tate and ate pairings, respectively, and $L = l_{[x^2]Q, [np^2]Q}([n]P)$. It follows from the bilinearities of $t_r(Q, P)$ and $a_p(Q, P)$ that

$$sopt(Q, P) = \left(f_{x, Q}^x([n]P) f_{x, \sigma(Q)}^p([n]P) f_{-2ax, Q}^p([n]P) f_{n, Q}^{p^2}([n]P) \cdot L \right)^{\frac{p^k-1}{r}} \quad (8)$$

also defines a bilinear pairing. By Lemma 1, one can prove that

$$\begin{aligned} f_{x, \sigma(Q)}^p([n]P) &= f_{x, \sigma(Q)}^p(\sigma \circ \hat{\sigma}(P)) \\ &= \left(f_{x, Q}^n(\hat{\sigma}(P)) \prod_{T \in \ker(\sigma) \setminus \{\mathcal{O}_E\}} \frac{corr_{[x]Q, T}(\hat{\sigma}(P))}{corr_{Q, T}^x(\hat{\sigma}(P))} \right)^p. \end{aligned}$$

Substituting this relation into Eq. (8), we obtain the final formula. Since $\sqrt{-D} \notin \mathbb{Q}(\zeta_k)$, as well as $a \neq 0$ or $k \not\equiv 0 \pmod{4}$, by [42, Theorem 2.4] we have

$$\deg(r(x)) = [\mathbb{Q}((a + b\sqrt{-D}) \cdot \zeta_k) : \mathbb{Q}] = [\mathbb{Q}(\zeta_k, \sqrt{-D}) : \mathbb{Q}] = 2\varphi(k).$$

Consequently, the length of Miller loop is $|x| \approx r(x)^{\frac{1}{2\varphi(k)}}$, which implies that this pairing is super-optimal.

The pairing $\text{sopt}(Q, P)$ degenerates when $t_n(Q, P)^{n\ell} = a_p(Q, P)^{n^2p-2anx}$. According to [40, Theorem 4], we obtain

$$t_r(Q, P)^{n\ell} = a_p(Q, P)^{n\ell kp^{k-1}((p^k-1)/r)^{-1} \bmod r}.$$

Hence, $\text{sopt}(Q, P)$ is nondegenerate if $\ell kp^{k-1} \not\equiv ((p^k-1)/r) \cdot (np-2ax) \bmod r$. This ends the proof. $\square$

In practice, we often expect that E admits a rational n -torsion point for efficient computations of the endomorphisms. This also benefits the implementation. Besides, it is worth noting that as the degree n increases, the corrections in Theorem 1 and Proposition 2 will be more complicated, potentially leading to expensive computations. Consequently, we commonly consider employing small degree endomorphisms, such as degrees $n = 2, 3$ for the constructions of super-optimal ate pairings.

3.2 Using degrees-2,3 endomorphisms to construct the super-optimal pairings

In this subsection, we will investigate the families admitting degrees-2,3 endomorphisms, and present concrete formulas for the super-optimal ate pairings.

Let $\sigma = a + b\sqrt{-D}$, $a, b \in \frac{1}{2}\mathbb{Z}$ denote a degree- n endomorphism. We first consider the case $n = 2$. If $D \equiv 1, 2 \bmod 4$, from Section 2.1 the endomorphism σ can be represented as $\sigma = a + b\sqrt{-D}$, $a, b \in \mathbb{Z}$, and it follows that

$$N(\sigma) = \sigma\hat{\sigma} = a^2 + b^2D = 2.$$

There are two possible integer solutions for (a^2, b^2D)

$$\begin{cases} a^2 = 0, & b^2D = 2, \\ a^2 = 1, & b^2D = 1, \end{cases}$$

which implies that

$$a^2 = b^2 = 1, D = 1 \text{ or } a^2 = 0, b^2 = 1, D = 2.$$

The curves with $D = 1$ have j -invariant 1728 [15]. Such curves possess efficiently computable automorphisms which can be leveraged to accelerate the pairing computations [11, 12, 34]. If $D = 2$, then the endomorphism σ can be represented as $\sigma = \pm\sqrt{-2}$.

When $D \equiv 3 \bmod 4$, then σ is of the form $\sigma = \frac{a'}{2} + \frac{b'\sqrt{-D}}{2}$, $a', b' \in \mathbb{Z}$, where $a = a'/2$, $b = b'/2$. It can be derived that

$$N(\sigma) = \sigma\hat{\sigma} = \frac{a'^2}{4} + \frac{b'^2D}{4} = 2,$$

i.e. $a'^2 + b'^2 D = 8$. Similarly, due to the fact that $D \in \mathbb{N}$ and $D \equiv 3 \pmod{4}$, there is only one possible choice

$$a'^2 = 1, \quad b'^2 D = 7,$$

which corresponds to $a' = \pm 1$, $b' = \pm 1$, $D = 7$. Hence, σ can be represented as $\sigma = \pm \frac{1}{2} \pm \frac{\sqrt{-7}}{2}$. In conclusion, only the curves with $D = 2$ and 7 possess degree-2 endomorphisms (also see Examples 5 and 6 in [19]). Furthermore, there exist no non-trivial automorphisms on these curves. Therefore, we can utilize the degree-2 endomorphisms for the constructions of super-optimal ate pairings.

We now concentrate on specific families $(p(x), t(x), r(x))$ of pairing-friendly curves with embedding degree k and $D = 2, 7$, and construct the super-optimal ate pairings. Based on the above analysis, families $F(2, 7, k, \pm \frac{1}{2}, \pm \frac{1}{2})$ and $F(2, 2, k, 0, \pm 1)$, which are respectively constructed by setting $r(x)$ as the minimal polynomial of $\alpha = \left(\frac{\pm 1 \pm \sqrt{-7}}{2}\right) \cdot \zeta_k$ and $\alpha = (\pm \sqrt{-2}) \cdot \zeta_k$, are compatible with the optimization techniques in Proposition 2.

Let $p(x), t(x), r(x)$ denote the polynomials parametrizing family $F(2, 7, k, \pm \frac{1}{2}, \pm \frac{1}{2})$ equipped with the degree-2 endomorphism $\sigma = (\pm 1 \pm \sqrt{-7})/2$. By Proposition 1 we have

$$\sigma \circ \pi(Q) = [x]Q \text{ and } x^2 \mp xp(x) + 2p(x)^2 \equiv 0 \pmod{r(x)}$$

for any $Q \in \mathbb{G}_2$. Thus, one of the short vectors $(c_0, \dots, c_{\varphi(k)-1})$ for the optimal ate pairing on family $F(2, 7, k, \pm \frac{1}{2}, \pm \frac{1}{2})$ is given by $(x^2, \mp x, 2, 0, \dots, 0)$. Similarly, the short vector for the optimal pairing on $F(2, 2, k, 0, \pm 1)$ is $(x^2, 0, 2, 0, \dots, 0)$. One can derive the super-optimal ate pairings on $F(2, 7, k, \pm \frac{1}{2}, \pm \frac{1}{2})$ and $F(2, 2, k, 0, \pm 1)$ by Proposition 2. For simplicity, we only provide the formulas for $F(2, 7, k, -\frac{1}{2}, \frac{1}{2})$ and $F(2, 2, k, 0, 1)$, which is summarized in Proposition 3.

Proposition 3. *Let the notations be denoted as above. Let E be a pairing-friendly curve in family $F(2, 2, k, 0, 1)$ or $F(2, 7, k, -\frac{1}{2}, \frac{1}{2})$ with $D = 2$ or 7 . Assume that $\sqrt{-2}, \sqrt{-7} \notin \mathbb{Q}(\zeta_k)$, as well as $k \not\equiv 0 \pmod{4}$. Define T_2 to be a 2-torsion point on E . Let $P \in \mathbb{G}_1$, $Q \in \mathbb{G}_2$. Then the formula for the super-optimal ate pairing $\mathbb{G}_2 \times \mathbb{G}_1 \rightarrow \mu_r$ on E is given by*

$$\text{sopt}(Q, P) = \begin{cases} \left(f_{x,Q}^x([2]P) \cdot f_{x,Q}^{2p}(\hat{\sigma}(P)) \cdot \text{CORR}(P) \right)^{\frac{p^k-1}{r}}, & E \in F(2, 2, k, 0, 1), \\ \left(f_{x,Q}^{x+p}([2]P) \cdot f_{x,Q}^{2p}(\hat{\sigma}(P)) \cdot \text{CORR}(P) \right)^{\frac{p^k-1}{r}}, & E \in F(2, 7, k, -\frac{1}{2}, \frac{1}{2}), \end{cases}$$

where $\hat{\sigma}$ denotes the conjugate element of σ . The correction is

$$\text{CORR}(P) = \begin{cases} \left(\frac{l_{Q,Q}^p([2]P) \cdot l_{[x]Q, T_2}(\hat{\sigma}(P)) \cdot \nu_{Q+T_2}^x(\hat{\sigma}(P))}{l_{Q, T_2}^x(\hat{\sigma}(P)) \cdot \nu_{[x]Q+T_2}(\hat{\sigma}(P))} \right)^p, & E \in F(2, 2, k, 0, 1), \\ \left(\frac{l_{\sigma(Q), [2]Q}^p([2]P) \cdot l_{Q,Q}^p([2]P) \cdot l_{[x]Q, T_2}(\hat{\sigma}(P)) \cdot \nu_{Q+T_2}^x(\hat{\sigma}(P))}{\nu_{[2]Q}^p([2]P) \cdot l_{Q, T_2}^x(\hat{\sigma}(P)) \cdot \nu_{[x]Q+T_2}(\hat{\sigma}(P))} \right)^p, & E \in F(2, 7, k, -\frac{1}{2}, \frac{1}{2}). \end{cases}$$

Besides, the pairing $\text{sopt}(Q, P)$ is nondegenerate if

$$\begin{cases} \ell k p^{k-1} \not\equiv ((p^k - 1)/r) \cdot 2p \pmod{r}, & E \in F(2, 2, k, 0, 1), \\ \ell k p^{k-1} \not\equiv ((p^k - 1)/r) \cdot (2p + x) \pmod{r}, & E \in F(2, 7, k, -\frac{1}{2}, \frac{1}{2}), \end{cases}$$

where ℓ is an integer such that $x^2 + xp + 2p^2 = \ell r$.

Proof. By Proposition 2, the formula of super-optimal ate pairing $\text{sopt}(Q, P)$ is given by

$$\left(f_{x,Q}^x([2]P) \cdot f_{x,Q}^{2p}(\hat{\sigma}(P)) \cdot f_{-2ax,Q}^p([2]P) \cdot f_{2,Q}^{p^2}([2]P) \cdot \text{CORR}(P) \right)^{\frac{p^k-1}{r}}, \quad (9)$$

where the degree-2 endomorphism $\sigma = a + b\sqrt{-D}$. The correction is

$$\text{CORR}(P) = l_{[x^2]Q, [2p^2]Q}([2]P) \cdot \left(\frac{\text{corr}_{[x]Q, T_2}(\hat{\sigma}(P))}{\text{corr}_{Q, T_2}^x(\hat{\sigma}(P))} \right)^p.$$

If $E \in F(2, 2, k, 0, 1)$, then $a = 0$, $b = 1$, and thus the function $f_{-2ax,Q}^p([2]P)$ vanishes. Since $x^2 + 2p^2 \equiv 0 \pmod{r}$, it holds that

$$l_{[x^2]Q, [2p^2]Q}([2]P) = \nu_{[2p^2]Q}([2]P) = \nu_{[2]Q}^{p^2}([2]P).$$

Additionally, it follows from the definitions of correction and Miller function that

$$f_{2,Q}^{p^2}([n]P) \cdot \text{CORR}(P) = \left(\frac{l_{Q,Q}^p([2]P) \cdot l_{[x]Q, T_2}(\hat{\sigma}(P)) \cdot \nu_{Q+T_2}^x(\hat{\sigma}(P))}{l_{Q, T_2}^x(\hat{\sigma}(P)) \cdot \nu_{[x]Q+T_2}(\hat{\sigma}(P))} \right)^p.$$

Plugging the above equality into Eq. (9), we obtain the formula for sopt on $E \in F(2, 2, k, 0, 1)$. As $\sqrt{-2} \notin \mathbb{Q}(\zeta_k)$ and $k \not\equiv 0 \pmod{4}$, by [42, Theorem 2.4] we have

$$\deg(r(x)) = [\mathbb{Q}(\zeta_k, \sqrt{-2}) : \mathbb{Q}] = 2\varphi(k),$$

i.e. $|x| \approx r(x)^{\frac{1}{2\varphi(k)}}$. Hence, $\text{sopt}(Q, P)$ is super-optimal. Besides, from Proposition 2 it can be deduced that $\text{sopt}(Q, P)$ is nondegenerate when

$$\ell k p^{k-1} \not\equiv ((p^k - 1)/r) \cdot 2p \pmod{r}.$$

The proof for $E \in F(2, 7, k, -\frac{1}{2}, \frac{1}{2})$ is similar so we omit it for simplicity. $\square$

Compared to the optimal ate pairing

$$\text{opt}(Q, P) = \left(f_{x^2,Q}(P) \cdot l_{Q,Q}^{p^2}(P) \right)^{\frac{p^k-1}{r}}$$

on $E \in F(2, 2, k, 0, 1)$, or

$$\text{opt}(Q, P) = \left(f_{x^2,Q}(P) \cdot f_{x,Q}^p(P) \cdot \frac{l_{[x^2]Q, [xp]Q}(P) l_{Q,Q}^{p^2}(P)}{\nu_{[2]Q}^{p^2}(P)} \right)^{\frac{p^k-1}{r}}$$

on $E \in F(2, 7, k, -\frac{1}{2}, \frac{1}{2})$, the number of basic Miller iterations for our new super-optimal ate pairing $sopt(Q, P)$ is shortened to $\lfloor \log_2(x) \rfloor \approx \log_2(r(x))/(2\varphi(k))$. Moreover, we can evaluate $f_{x,Q}([2]P)$ and $f_{x,Q}(\hat{\sigma}(P))$ in a shared Miller loop, which further speeds up the pairing computation. For efficiency purpose, we also expect that $2 \mid \#E(\mathbb{F}_p)$ to ensure that E possesses an $\mathbb{F}_p$ -rational 2-torsion point T_2 . In Section 4, we will explore how to efficiently implement the pairing computation on GG22D7 using our new formulas in detail.

Remark 1. As mentioned before, our approach is not limited to be applicable for family $F(2, 7, k, -\frac{1}{2}, \frac{1}{2})$. Any family of the pairing-friendly curves constructed by making $r(x)$ the minimal polynomial of $\alpha = (\frac{\pm 1 \pm \sqrt{-7}}{2}) \cdot \zeta_k$ with $\sqrt{-7} \notin \mathbb{Q}(\zeta_k)$, or $\alpha = (\pm \sqrt{-2}) \cdot \zeta_k$ with $\sqrt{-2} \notin \mathbb{Q}(\zeta_k)$ is tailored to the acceleration techniques in Proposition 3.

Application on family GG22D7. As mentioned in Section 1, Gasnier and Guillevic proposed an excellent family of pairing-friendly curves named GG22D7 with $D = 7$ [20] and embedding degree $k = 22$. This family possesses several specific features such as small ρ -value. A curve named GG22D7-457 in this family benefits from its efficient $\mathbb{G}_1$ arithmetic due to the small characteristic p .

Example 1. (GG22D7, [20]) Let $k = 22$, $D = 7$, $\sigma = (-1 + \sqrt{-7})/2$, $\alpha = \sigma \cdot \zeta_k$.

$$\begin{aligned} t(x) &= (x^{12} + 45x + 46)/46, \\ p(x) &= (x^{24} - x^{23} + 2x^{22} + 67x^{13} + 94x^{12} + 134x^{11} + 2048x^2 + 4096)/7406, \\ r(x) &= (x^{20} - x^{19} - x^{18} + 3x^{17} - x^{16} - 5x^{15} + 7x^{14} + 3x^{13} - 17x^{12} + 11x^{11} \\ &\quad + 23x^{10} + 22x^9 - 68x^8 + 24x^7 + 112x^6 - 160x^5 - 64x^4 + 384x^3 \\ &\quad - 256x^2 - 512x + 1024)/23. \end{aligned}$$

We conclude that $\text{GG22D7} \in F(2, 7, 22, -\frac{1}{2}, \frac{1}{2})$ satisfies the conditions in Proposition 3, and thus is compatible with our improvement approach. In other words, the pairing computation on this family can be optimized by utilizing the degree-2 endomorphism σ . The explicit formula for the super-optimal ate pairing on GG22D7 is presented in Proposition 3.

We now focus on the utilization of degree-3 endomorphism. Let σ denote a degree-3 endomorphism. If $D \equiv 1, 2 \pmod{4}$, then we can represent σ as $\sigma = a + b\sqrt{-D}$, $a, b \in \mathbb{Z}$, and it holds that

$$a^2 + b^2 D = 3,$$

which implies that $a^2 = 1$, $b^2 D = 2$. It corresponds to curves with $D = 2$, whose pairing computation can be enhanced by employing degree-2 endomorphisms based on the analysis above. Therefore, integrating degree-3 endomorphisms into the such curves is unnecessary.

If $D \equiv 3 \pmod{4}$, then it yields that $\sigma = \frac{a'}{2} + \frac{b'\sqrt{-D}}{2}$, $a', b' \in \mathbb{Z}$ and $a = a'/2$, $b = b'/2$. We deduce that

$$\sigma\hat{\sigma} = \left(\frac{a'}{2} + \frac{b'\sqrt{-D}}{2}\right) \left(\frac{a'}{2} - \frac{b'\sqrt{-D}}{2}\right) = 3,$$

which implies that

$$a'^2 + b'^2 D = 12.$$

By $D \equiv 3 \pmod{4}$ and $D \in \mathbb{N}$, there are the following three possible solutions

$$\begin{cases} a'^2 = 0, & b'^2 D = 12, \\ a'^2 = 1, & b'^2 D = 11, \\ a'^2 = 9, & b'^2 D = 3. \end{cases}$$

It is worth noting that except for the second situation, the corresponding CM-discriminant is $D = 3$, which corresponds to the curves admitting efficiently computable automorphisms. Therefore, we only focus on the case with respect to $D = 11$, where σ can be expressed as $\sigma = \pm\frac{1}{2} \pm \frac{\sqrt{-11}}{2}$.

Similar to the case $n = 2$, we utilize the techniques in Proposition 2 to families $F(3, 11, k, \pm\frac{1}{2}, \pm\frac{1}{2})$, which are constructed by making $r(x)$ the minimal polynomial of $\alpha = (\frac{\pm 1 \pm \sqrt{-11}}{2}) \cdot \zeta_k$. We set $\sigma = \frac{-1 + \sqrt{-11}}{2}$ for simplicity and show how to exploit the GLV-endomorphism σ to enhance the efficiency of the pairing computations for $F(3, 11, k, -\frac{1}{2}, \frac{1}{2})$. By Proposition 1 we have

$$x^2 + xp(x) + 3p(x)^2 \equiv 0 \pmod{r(x)}.$$

Consequently, one of the short vectors $(c_0, \dots, c_{\varphi(k)-1})$ for the optimal pairing on $F(3, 11, k, -\frac{1}{2}, \frac{1}{2})$ is given by $(x^2, x, 3, 0, \dots, 0)$. Our new formula for the super-optimal pairing on $F(3, 11, k, -\frac{1}{2}, \frac{1}{2})$ leveraging degree-3 endomorphism σ is stated in the following proposition.

Proposition 4. *Let the notation be denoted as above. Let E be a pairing-friendly curve in $F(3, 11, k, -\frac{1}{2}, \frac{1}{2})$ with $D = 3$. Assume that $\sqrt{-3} \notin \mathbb{Q}(\zeta_k)$, together with $a \neq 0$ or $k \not\equiv 0 \pmod{4}$. Let $P \in \mathbb{G}_1$, $Q \in \mathbb{G}_2$. Define $T_3 \in E(\mathbb{F}_p)$ to be a generator of $\ker(\sigma)$. Then the formula of the super-optimal ate pairing $\text{sopt} : \mathbb{G}_2 \times \mathbb{G}_1 \rightarrow \mu_r$ for on $E \in F(3, 11, k, -\frac{1}{2}, \frac{1}{2})$ is given as follows*

$$\text{sopt}(Q, P) = \left(f_{x,Q}^{x+p}([3]P) \cdot f_{x,Q}^{3p}(\hat{\sigma}(P)) \cdot \text{CORR}(P) \right)^{\frac{p^k-1}{r}},$$

where $\hat{\sigma}$ is the dual of σ . The correction $\text{CORR}(P)$ is

$$\text{CORR}(P) = \left(\frac{l_{\sigma(Q), [3]Q}^p([3]P) \cdot f_{3,Q}^p([3]P) \cdot \text{corr}_{[x]Q, T_3}(\hat{\sigma}(P)) \cdot \text{corr}_{[x]Q, -T_3}(\hat{\sigma}(P))}{\text{corr}_{Q, T_3}^x(\hat{\sigma}(P)) \cdot \text{corr}_{Q, -T_3}^x(\hat{\sigma}(P))} \right)^p.$$

Besides, the pairing $\text{sopt}(Q, P)$ is nondegenerate if

$$\ell p^{k-1} \not\equiv ((p^k - 1)/r) \cdot (3p + x) \pmod{r},$$

where ℓ denotes an integer such that $x^2 + xp + 3p^2 = \ell r$.

Proof. The proof is almost the same as Proposition 3. By invoking Proposition 2 and using the fact that $[2]T_3 = -T_3$, one can derive the formula for the super-optimal ate pairing on $F(3, 11, k, -\frac{1}{2}, \frac{1}{2})$.

Compared with the optimal ate pairing

$$\text{opt}(Q, P) = \left(f_{x^2, Q}(P) \cdot f_{x, Q}^p(P) \cdot l_{[x^2]Q, [xp]Q}(P) \cdot f_{3, Q}^{p^2}(P) \right)^{\frac{p^k - 1}{r}},$$

the bit-length of Miller loop of $\text{sopt}(Q, P)$ achieves about $\log_2(r(x))/(2\varphi(k))$ and we can simultaneously evaluate $f_{x, Q}([3]P)$ and $f_{x, Q}(\hat{\sigma}(P))$ in a shared Miller loop. Besides, the computation of correction $\text{CORR}(P)$ is relatively inexpensive. Therefore, exploiting degree-3 endomorphisms can crucially enhance the performance of the pairing computation.

Application on family GG28D11. As an illustrative example, we present the following family constructed by Gasnier and Guillevis named GG28D11 [20] with embedding degree $k = 28$.

Example 2. (GG28D11 [20]) $k = 28$, $D = 11$, $\sigma = (-1 + \sqrt{-11})/2$, $\alpha = \sigma \cdot \zeta_k$.

$$\begin{aligned} t(x) &= (x^{15} + 718x + 3237)/3237, \\ p(x) &= (x^{30} + x^{29} + 3x^{28} + 2515x^{16} + 14384x^{15} + 7545x^{14} + 4782969x^2 \\ &\quad + 13304911x + 14348907)/38419953, \\ r(x) &= (x^{24} + 5x^{22} + 16x^{20} + 35x^{18} + 31x^{16} - 160x^{14} - 1079x^{12} - 1440x^{10} \\ &\quad + 2511x^8 + 25515x^6 + 104976x^4 + 295245x^2 + 531441). \end{aligned}$$

It yields that $\text{GG28D11} \in F(3, 11, 28, -\frac{1}{2}, \frac{1}{2})$, which is compatible with our acceleration method. Thus the corresponding pairing computation can be sped up by employing degree-3 endomorphism σ . Concrete formula for the super-optimal ate pairing on this family is presented in Proposition 4.

4 Efficient implementation for the super-optimal pairing utilizing endomorphisms

In this section, we consider family GG22D7 for implementation, and employ the optimization techniques in Section 3 to accelerate the corresponding pairing computation. Detailed computational procedure of the shared Miller loop and concrete cost analysis for our new super-optimal pairing are also presented. Furthermore, we provide the experimental results of our optimized implementation.

4.1 Choice of parameters at the 192-bit security level

As established in [20], taking $x = -779523$ in family GG22D7 (Example 1) yields a pairing-friendly curve admitting a degree-2 endomorphism σ at the 192-bit security level. Gasnier and Guillevis [20] named this curve as GG22D7-457

to distinguish it by the embedding degree k , the CM-discriminant D and the characteristic p . Curve GG22D7-457 is over $\mathbb{F}_p$, possessing coefficient $a = -3$ and j -invariant -3375 [20]. Additionally, the extension field $\mathbb{F}_{p^{22}}$ can be constructed as follows:

$$\mathbb{F}_p - \mathbb{F}_{p^{11}} \cong \mathbb{F}_p[\xi]/(\xi^{11} - 2\xi - 2) - \mathbb{F}_{p^{22}} \cong \mathbb{F}_{p^{11}}[v]/(v^2 - \xi).$$

Based on the above analysis, we take the degree-2 endomorphism as $\sigma = \frac{-1+\sqrt{-7}}{2}$ and obtain

$$\tau(Q) = \sigma \circ \pi(Q) = [-x]Q.$$

It follows from Proposition 1 that

$$x^2 - xp(x) + 2p(x)^2 \equiv 0 \pmod{r(x)}.$$

By Proposition 3, our formula for the super-optimal ate pairing on GG22D7-457 is

$$\text{sopt}(Q, P) = \left(f_{-x, Q}^{p-x}([2]P) \cdot f_{-x, Q}^{2p}(\hat{\sigma}(P)) \cdot \text{CORR}(P) \right)^{\frac{p^{22}-1}{r}}, \quad (10)$$

where

$$\text{CORR}(P) = \left(\frac{l_{\sigma(Q), [2]Q}^p([2]P) \cdot l_{Q, Q}^p([2]P) \cdot l_{[-x]Q, T_2}(\hat{\sigma}(P)) \cdot \nu_{Q+T_2}^{-x}(\hat{\sigma}(P))}{\nu_{[2]Q}^p([2]P) \cdot l_{Q, T_2}^{-x}(\hat{\sigma}(P)) \cdot \nu_{[-x]Q+T_2}(\hat{\sigma}(P))} \right)^p.$$

Therefore, the bit-length of Miller loop is about

$$\log_2(|x|) \approx \frac{1}{22} \log_2(r(x)),$$

which achieves $\log_2(r(x))/2\varphi(k)$. We have also verified that $2 \mid \#E(\mathbb{F}_p)$ i.e. $T_2 \in E(\mathbb{F}_p)[2]$.

4.2 Shared Miller loop

In this subsection, we present explicit computational procedure for the doubling and addition steps in a shared Miller loop. The notations for cost analysis in the remaining part of this paper are first introduced.

Notations. Let $\mathbf{m}$, $\mathbf{s}$, and $\mathbf{i}$ denote the computational costs of a multiplication, a squaring and an inversion in $\mathbb{F}_p$, respectively. Let $\mathbf{m}_{11}$, $\mathbf{s}_{11}$, $\mathbf{i}_{11}$ and $\mathbf{f}_{11}$ represent the overheads of a multiplication, a squaring, an inversion and a Frobenius endomorphism in $\mathbb{F}_{p^{11}}$, respectively. Denote by $\mathbf{m}_{22}$, $\mathbf{s}_{22}$, $\mathbf{i}_{22}$, $\mathbf{f}_{22}$ and $\mathbf{e}_x$ the costs of a multiplication, a squaring, an inversion, a Frobenius endomorphism and an exponentiation by $|x|$ in $\mathbb{F}_{p^{22}}$. We omit the calculation of the additions over finite fields for simplicity.

Recalled from Eq. (10), in the computational phase of $\text{sopt}(Q, P)$ on GG22D7-457, we need to evaluate two Miller functions $f_{x, Q}(\hat{\sigma}(P))$ and $f_{x, Q}([2]P)$. Computing a single pairing with multiple evaluations of Miller functions were studied

in [3,12,43]. Inspired by these works, we simultaneously accomplished these two evaluations in a shared Miller loop.

Since curve GG22D7-457 has the form $E : y^2 = x^3 + ax + b$ with embedding degree $k = 22$ and CM-discriminant $D = 7$, it admits a quadratic twist $E' : y^2 = x^3 + a/\xi^2 x + b/\xi^3$ [23] over $\mathbb{F}_{p^{11}}$. The twisting isomorphism is given by

$$\phi : (x, y) \mapsto (x\xi, y\xi v).$$

Thanks to this twisting map, the subgroup $\mathbb{G}_2$ can be represented as

$$\mathbb{G}_2 = E(\mathbb{F}_{p^{22}})[r] \cap \ker(\pi - [p]) \cong E'(\mathbb{F}_{p^{11}})[r].$$

We work with Jacobian coordinates for any point $P = (x_P, y_P) \in E$ to avoid inversion, i.e. the point P is represented by $P = (X_P, Y_P, Z_P)$ in projective space, where $x_P = X_P/Z_P^2$, $y_P = Y_P/Z_P^3$. The denominator elimination can be leveraged since the embedding degree is even. Consequently, it is sufficient to update the numerator. The function $f_{1,Q}$ can be initialized as

$$f_{1,Q}([2]P) = f_{1,Q}(\hat{\sigma}(P)) = 1.$$

The Jacobian coordinates of $[2]P$ and $\hat{\sigma}(P)$ can be derived using the mixed doubling [6] and Vélú's formulas [38], respectively. We omit the computational procedures for simplicity. The costs of evaluating $[2]$ and $\hat{\sigma}$ at P are $1\mathbf{m}+5\mathbf{s}$ and $5\mathbf{m}+1\mathbf{s}$, respectively. Subsequently, we compute the affine coordinates $x_{[2]P}, y_{[2]P}, x_{\hat{\sigma}(P)}$ and $y_{\hat{\sigma}(P)}$ employing the batch inversion at a cost of $9\mathbf{m}+2\mathbf{s}+1\mathbf{i}$. Therefore, the cost of initialization is $15\mathbf{m}+8\mathbf{s}+1\mathbf{i}$. In the following, we explore how to update the intermediate values $f_{m,Q}([2]P)$ and $f_{m,Q}(\hat{\sigma}(P))$ in the shared Miller loop.

Shared addition step. Let $Q' = (x_{Q'}, y_{Q'}) \in E'(\mathbb{F}_{p^{11}})[r]$. Denote the Jacobian coordinates of $[m]Q'$ by (X_T, Y_T, Z_T) . We adopt the mixed addition formula in [6] to obtain the point $[m+1]Q'$, which can be executed by the following sequences:

$$\begin{aligned} ZZ &\leftarrow Z_T^2, \quad U \leftarrow x'_{Q'} \cdot ZZ, \quad S \leftarrow y'_{Q'} \cdot Z_T \cdot ZZ, \quad H \leftarrow U - X_T, \\ I &\leftarrow (2H)^2, \quad J \leftarrow H \cdot I, \quad \alpha_{T+Q'} \leftarrow S - Y_T, \quad W \leftarrow 2\alpha_{T+Q'}, \quad V \leftarrow X_T \cdot I, \\ X_{T+Q'} &\leftarrow W^2 - J - 2V, \quad Y_{T+Q'} \leftarrow W \cdot (V - X_{T+Q'}) - 2Y_T \cdot J, \\ Z_{TQ'} &\leftarrow Z_T \cdot H, \quad Z_{T+Q'} \leftarrow 2Z_{TQ'}. \end{aligned}$$

The above operations achieve a computational cost of $8\mathbf{m}_{11}+3\mathbf{s}_{11}$. Acting the twisting isomorphism ϕ on T and Q' yields

$$\phi(T) = (X_T\xi, Y_T\xi v, Z_T), \quad Q = \phi(Q') = (x_{Q'}\xi, y_{Q'}\xi v) \in \mathbb{G}_2.$$

It follows from the iteration formula of $f_{m,Q}$ that

$$f_{m+1,Q}(x, y) \leftarrow f_{m,Q}(x, y) \cdot L_{\phi(T), \phi(Q')}(x, y)$$

where the function $L_{\phi(T),\phi(Q')}(x, y)$ is defined as follows:

$$L_{\phi(T),\phi(Q')}(x, y) = Z_{TQ'} \cdot y - (Z_{TQ'} \cdot y_{Q'}\xi + \alpha_{T+Q'}(x - x_{Q'}\xi))v.$$

It is worth noting that $\alpha_{T+Q'}$ and $Z_{TQ'}$ have been derived after computing $T + Q$. Furthermore, the computation of multiplying an element of $\mathbb{F}_{p^{11}}$ by ξ only requires several additions. Therefore, we perform the following sequence to compute $L_{\phi(T),\phi(Q')}([2]P)$ and $L_{\phi(T),\phi(Q')}(\hat{\sigma}(P))$

$$\begin{aligned} B &\leftarrow Z_{TQ'} \cdot y_{Q'}\xi, \quad C \leftarrow \alpha_{T+Q'} \cdot x_{Q'}\xi, \quad D \leftarrow B - C + \alpha_{T+Q'}x_{[2]P}, \\ E &\leftarrow B - C + \alpha_{T+Q'}x_{\hat{\sigma}(P)}, \quad L_{\phi(T),\phi(Q')}([2]P) \leftarrow Z_{TQ'} \cdot y_{[2]P} - Dv, \\ L_{\phi(T),\phi(Q')}(\hat{\sigma}(P)) &\leftarrow Z_{TQ'} \cdot y_{\hat{\sigma}(P)} - Ev \end{aligned}$$

at a cost of $2\mathbf{m}_{11} + 44\mathbf{m}$. Finally, we execute $2\mathbf{m}_{22}$ to obtain

$$\begin{aligned} f_{m+1,Q}([2]P) &\leftarrow f_{m,Q}([2]P) \cdot L_{\phi(T),\phi(Q')}([2]P), \\ f_{m+1,Q}(\hat{\sigma}(P)) &\leftarrow f_{m,Q}(\hat{\sigma}(P)) \cdot L_{\phi(T),\phi(Q')}(\hat{\sigma}(P)). \end{aligned}$$

Consequently, the total computational cost of a shared addition step is

$$\begin{aligned} \text{Cost}_{\text{SADD}} &= 8\mathbf{m}_{11} + 3\mathbf{s}_{11} + 2\mathbf{m}_{11} + 44\mathbf{m} + 2\mathbf{m}_{22} \\ &= 2\mathbf{m}_{22} + 10\mathbf{m}_{11} + 3\mathbf{s}_{11} + 44\mathbf{m}. \end{aligned}$$

Shared doubling step. We first leverage the doubling formula for $a = -3$ [6] to derive $[2]T$, which can be expressed as the following sequence of operations

$$\begin{aligned} YY &= Y_T^2, \quad ZZ \leftarrow Z_T^2, \quad S = X_T \cdot YY, \quad \beta_{[2]T} \leftarrow 3(X_T - ZZ/\xi) \cdot (X_T + ZZ/\xi), \\ N &\leftarrow \beta_{[2]T}^2 - 8S, \quad X_{[2]T} \leftarrow N, \quad Y_{[2]T} \leftarrow \beta_{[2]T} \cdot (4S - X_{[2]T}) - 8YY^2, \\ Z_{[2]T} &\leftarrow (Y_T + Z_T)^2 - YY - ZZ. \end{aligned}$$

From the above sequence, the cost for performing a point doubling is $3\mathbf{m}_{11} + 5\mathbf{s}_{11}$. We can update $f_{m,Q}$ by

$$f_{2m,Q}(x, y) \leftarrow f_{m,Q}^2(x, y) \cdot L_{\phi(T),\phi(T)}(x, y),$$

where the line function $L_{\phi(T),\phi(T)}(x, y)$ is given by

$$L_{\phi(T),\phi(T)}(x, y) = Z_{[2]T}Z_T^2y - (2Y_T^2\xi + (xZ_T^2 - X_T\xi) \cdot \beta_{[2]T})v.$$

Note that the values $Z_{[2]T}, \beta_{[2]T}, Y_T^2$ and Z_T^2 have been obtained during the calculations of $[2]T$. Consequently, we can execute the following sequence to compute $L_{\phi(T),\phi(T)}([2]P)$ and $L_{\phi(T),\phi(T)}(\hat{\sigma}(P))$

$$\begin{aligned} B &\leftarrow 2Y_T^2\xi, \quad C \leftarrow \beta_{[2]T}Z_T^2, \quad D \leftarrow \beta_{[2]T}X_T, \quad E \leftarrow Z_{[2]T}Z_T^2, \\ F &\leftarrow E \cdot y_{[2]P}, \quad G \leftarrow E \cdot y_{\hat{\sigma}(P)}, \quad L_{\phi(T),\phi(T)}([2]P) \leftarrow F - (B + Cx_{[2]P} - D\xi)v, \\ L_{\phi(T),\phi(T)}(\hat{\sigma}(P)) &\leftarrow G - (B + Cx_{\hat{\sigma}(P)} - D\xi)v, \end{aligned}$$

which requires $3\mathbf{m}_{11} + 44\mathbf{m}$.

Finally, updating $L_{\phi(T),\phi(T)}([2]P)$ and $L_{\phi(T),\phi(T)}(\hat{\sigma}(P))$ yields $2\mathbf{m}_{22} + 2\mathbf{s}_{22}$. Hence, the total cost of a shared doubling step is

$$\begin{aligned}\text{Cost}_{\text{SDBL}} &= 3\mathbf{m}_{11} + 5\mathbf{s}_{11} + 3\mathbf{m}_{11} + 44\mathbf{m} + 2\mathbf{m}_{22} + 2\mathbf{s}_{22} \\ &= 2\mathbf{m}_{22} + 2\mathbf{s}_{22} + 6\mathbf{m}_{11} + 5\mathbf{s}_{11} + 44\mathbf{m}.\end{aligned}$$

The first doubling step is a bit special, since we can use the affine coordinate $T = Q' = (x_{Q'}, y_{Q'})$ and employ the mixed doubling formula [6] to compute $[2]T$. The corresponding cost is

$$\text{Cost}_{\text{SDBL}_1} = 3\mathbf{m}_{11} + 5\mathbf{s}_{11} + 22\mathbf{m}.$$

4.3 Cost analysis and comparison

In this subsection, we provide the computational cost analysis for the Miller loop of super-optimal ate pairing on GG22D7-457, and compare it with the optimal ate pairing in [20].

As $\gcd(p^{21}, r) = 1$, the power $\text{sopt}(Q, P)^{p^{21}}$ also defines a nondegenerate bilinear pairing. Besides, one can show that

$$\nu_{Q+T_2}(\hat{\sigma}(P)), \nu_{[-x]Q+T_2}(\hat{\sigma}(P)) \text{ and } \nu_{[2]Q}([2]P) \in \mathbb{F}_{p^{11}},$$

which can be killed by the final exponentiation. Recalled from Eq. (10), the super-optimal ate pairing $\text{sopt}(Q, P)^{p^{21}}$ can be re-expressed by

$$\left(\left(f_{-x,Q}^{p^{21}}([2]P) / l_{Q,T_2}(\hat{\sigma}(P)) \right)^{\frac{-x+1}{2}} \cdot f_{-x,Q}(\hat{\sigma}(P)) \right)^2 \cdot \frac{M_1}{M_2},$$

where

$$\begin{aligned}M_1 &= (l_{\sigma(Q),[2]Q}([2]P) \cdot l_{Q,Q}([2]P) \cdot l_{\sigma(Q),T_2}(\hat{\sigma}(P)))^p \cdot f_{-x,Q}([2]P), \\ M_2 &= f_{-x,Q}^{p^{21}}([2]P) / l_{Q,T_2}(\hat{\sigma}(P)).\end{aligned}$$

Since the embedding degree $k = 22$ is even, the inversion of an element in $\mathbb{F}_{p^{22}}$ can be replaced by multiplying its conjugate. The 2-NAF representation can be applied to reduce the Hamming weight of the length of Miller loop. Thus $-x = 779523$ and the exponent $\frac{-x+1}{2} = 389762$ can be represented as

$$-x = [1 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ -1]_2$$

and

$$\frac{-x+1}{2} = [1 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0]_2,$$

respectively. We can simultaneously accomplish the exponentiation

$$\left(f_{-x,Q}^{p^{21}}([2]P) \cdot \overline{l_{Q,T_2}(\hat{\sigma}(P))} \right)^{\frac{-x+1}{2}}$$

and the evaluation of $f_{-x,Q}(\hat{\sigma}(P))$ by using the techniques in [21,35,11]. The shared Miller loop for the computation of

$$\left(f_{-x,Q}^{p^{21}}([2]P) \cdot \overline{l_{Q,T_2}(\hat{\sigma}(P))}\right)^{\frac{-x+1}{2}} \cdot f_{-x,Q}(\hat{\sigma}(P))$$

is illustrated in Algorithm 1. From Algorithm 1, it requires 6 shared addition steps, 20 shared doubling steps together with the extra cost of $8\mathbf{m}_{22} + 1\mathbf{f}_{22}$ to obtain $\left(f_{-x,Q}^{p^{21}}([2]P) \cdot \overline{l_{Q,T_2}(\hat{\sigma}(P))}\right)^{\frac{-x+1}{2}} \cdot f_{-x,Q}(\hat{\sigma}(P))$. The corresponding cost of the shared Miller loop is

$$\begin{aligned} \text{Cost}_{\text{Miller}} &= \text{Cost}_{\text{SDBL}_1} + 19\text{Cost}_{\text{SDBL}} + 6\text{Cost}_{\text{SADD}} + 8\mathbf{m}_{22} + 1\mathbf{f}_{22} \\ &= 58\mathbf{m}_{22} + 38\mathbf{s}_{22} + 177\mathbf{m}_{11} + 118\mathbf{s}_{11} + 1122\mathbf{m} + 1\mathbf{f}_{22}. \end{aligned}$$

Note that $l_{Q,Q}([2]P)$ have been obtained after executing the first shared doubling steps of $f_{-x,Q}(\hat{\sigma}(P))$ and $f_{-x,Q}([2]P)$, respectively. Computing the remaining three line functions $l_{\sigma(Q),[2]Q}([2]P)$, $l_{Q,T_2}(\hat{\sigma}(P))$ and $l_{\sigma(Q),T_2}(\hat{\sigma}(P))$ yields $6\mathbf{m}_{11} + 44\mathbf{m}$. We omit the cost analysis of these line functions for simplicity.

It remains to compute

$$\left(\left(f_{-x,Q}^{p^{21}}([2]P) \cdot \overline{l_{Q,T_2}(\hat{\sigma}(P))}\right)^{\frac{-x+1}{2}} \cdot f_{-x,Q}(\hat{\sigma}(P))\right)^2 \cdot \frac{M_1}{M_2},$$

The corresponding cost of the remaining accumulation is $4\mathbf{m}_{22} + 1\mathbf{s}_{22} + \mathbf{f}_{22}$. Consequently, the total cost of the Miller function evaluation is:

$$\begin{aligned} \text{Cost}_{\text{ML}} &= \text{Cost}_{\text{Init}} + \text{Cost}_{\text{Miller}} + \text{Cost}_{\text{Remain}} \\ &= 62\mathbf{m}_{22} + 39\mathbf{s}_{22} + 183\mathbf{m}_{11} + 118\mathbf{s}_{11} + 1181\mathbf{m} + 8\mathbf{s} + 2\mathbf{f}_{22} + 1\mathbf{i}. \end{aligned}$$

The relative cost of multiplication, squaring, Frobenius endomorphism, inversion and exponentiation in $\mathbb{F}_p$, $\mathbb{F}_{p^{11}}$ or $\mathbb{F}_{p^{22}}$ are presented in Table 1 [20]. As for the exponentiation, we utilize addition chain to estimate $\mathbf{e}_x$ in $\mathbb{F}_{p^k}$, which requires $6\mathbf{m}_k + 18\mathbf{s}_k$. By setting $\mathbf{s} = \mathbf{m}$, we can estimate the cost of the Miller loop and the

Table 1. Cost of multiplication, squaring, Frobenius endomorphism, inversion and exponentiation in $\mathbb{F}_{p^k}$ [20]. The inversion in base field $\mathbb{F}_p$ is estimated with $\mathbf{i} = 25\mathbf{m}$.

k	$\mathbf{m}_k$	$\mathbf{s}_k$	$\mathbf{f}_k$	$\mathbf{i}_k$	$\mathbf{e}_x$
1	$\mathbf{m}$	$\mathbf{s}$	0	$25\mathbf{m}$	$6\mathbf{m} + 18\mathbf{s}$
11	$48\mathbf{m}$ [33]	$48\mathbf{s}$	$110\mathbf{m}$	$789\mathbf{m}$	$288\mathbf{m} + 864\mathbf{s}$
22	$3\mathbf{m}_{11} = 144\mathbf{m}$	$2\mathbf{m}_{11} = 96\mathbf{m}$	$231\mathbf{m}$	$981\mathbf{m}$	$864\mathbf{m} + 1728\mathbf{s}$

final exponentiation. Table 2 shows the comparison of the computational cost

Algorithm 1 Shared Miller loop for computing $\left(f_{-x,Q}^{p^{21}}([2]P) \cdot \overline{l_{Q,T_2}(\hat{\sigma}(P))}\right)^{\frac{-x+1}{2}} \cdot f_{-x,Q}(\hat{\sigma}(P))$.

Input: Three points Q , $[2]P$, $\hat{\sigma}(P)$, the line function $l_{Q,T_2}(\hat{\sigma}(P))$ and the 2-NAF representation $-x = \sum_{i=0}^N n_i 2^i$ with $n_i \in \{-1, 0, 1\}$.

Output: The value $\left(f_{-x,Q}^{p^{21}}([2]P) \cdot \overline{l_{Q,T_2}(\hat{\sigma}(P))}\right)^{\frac{-x+1}{2}} \cdot f_{-x,Q}(\hat{\sigma}(P))$.

```

1:  $T \leftarrow Q$ ,  $f \leftarrow 1$ ,  $g \leftarrow 1$ ,  $tab \leftarrow []$ ,  $j \leftarrow 0$ 
2: for  $i = N - 1$  to 0 do
3:    $f \leftarrow f^2 \cdot l_{T,T}([2]P)$ ,  $tab[j] \leftarrow l_{T,T}(\hat{\sigma}(P))$ ,  $T \leftarrow [2]T$ ,  $j \leftarrow j + 1$ 
4:   if  $i = N - 1$  then
5:      $M_2 \leftarrow l_{T,T}([2]P)$ 
6:   end if
7:   if  $n_i = 1$  then
8:      $f \leftarrow f \cdot l_{T,Q}([2]P)$ ,  $tab[j] \leftarrow l_{T,Q}(\hat{\sigma}(P))$ ,  $T \leftarrow T + Q$ ,  $j \leftarrow j + 1$ 
9:   else if  $n_i = -1$  then
10:     $f \leftarrow f \cdot l_{T,-Q}([2]P)$ ,  $tab[j] \leftarrow l_{T,-Q}(\hat{\sigma}(P))$ ,  $T \leftarrow T - Q$ ,  $j \leftarrow j + 1$ 
11:  end if
12: end for                                     //  $f = f_{-x,Q}([2]P)$ 
13:  $j \leftarrow 0$ ,  $i \leftarrow N - 1$ 
14:  $g \leftarrow g^2 \cdot tab[j]$ ,  $j \leftarrow j + 1$ 
15: if  $n_i = 1$  or  $n_i = -1$  then
16:    $g \leftarrow g \cdot tab[j]$ ,  $j \leftarrow j + 1$ 
17: end if                                     //  $g = f_{2,Q}(\hat{\sigma}(P))$ 
18:  $h \leftarrow f^{p^{21}} \cdot \overline{l_{Q,T_2}(\hat{\sigma}(P))} \cdot g$ ,  $h_1 \leftarrow f^{p^{21}} \cdot \overline{l_{Q,T_2}(\hat{\sigma}(P))}$ 
19: for  $i = N - 2$  to 0 do
20:    $h \leftarrow h^2 \cdot tab[j]$ ,  $j \leftarrow j + 1$ 
21:   if  $n_i = 1$  or  $n_i = -1$  then
22:      $h \leftarrow h \cdot tab[j]$ ,  $j \leftarrow j + 1$ 
23:   end if
24:   if  $n_{i+1} = 1$  then
25:      $h \leftarrow h \cdot h_1$ ,  $j \leftarrow j + 1$ 
26:   else if  $n_{i+1} = -1$  then
27:      $h \leftarrow h \cdot h_1$ ,  $j \leftarrow j + 1$ 
28:   end if
29: end for
30: return  $h$ 

```

between the optimal ate pairing and our new pairing formula on GG22D7-457. For the final exponentiation, we refer to the SageMath code in [2,20] to derive the computational cost.

From Table 2 we can see that our new formula exploiting the degree-2 endomorphism outperforms the optimal ate pairing for the Miller loop on GG22D7-

Table 2. The cost of pairing computation exploiting optimal pairing and super-optimal pairing with degree-2 endomorphism on curve GG22D7-457, including Miller function evaluation and the final exponentiation.

	This work	Optimal pairing [2,20]	Ratio
Miller loop	28796m	42276m	68.1%
Final exp	73848m	73848m	-
Total	102644m	116124m	88.4%

457, obtaining a saving of **31.9%** $\mathbb{F}_p$ -multiplications. As for the total pairing computation, we also achieve a reduction of 11.6% in terms of $\mathbb{F}_p$ -multiplications.

4.4 Implementation results

Our implementation is based on a well-known cryptographic library RELIC [1]. We integrate our techniques into RELIC and obtain a performance comparison for pairing computations between GG22D7-457 and other well-known pairing-friendly curves at 192-bit security level. We compile and benchmark our code on Intel(R) Core(TM) i9-12900K 3.20 GHz with TurboBoost and hyperthreading features disabled. Table 3 illustrates the detail of the comparison.

According to Table 3, by utilizing the degree-2 endomorphism, the Miller loop for the super-optimal pairing is about **26.0%** faster than the optimal ate pairing on GG22D7-457. Compared to the other curves, the optimized Miller loop on GG22D7-457 is about 23.3%, 4.7% and 8.3% faster than that on curves KSS16-766, KSS18-638 and FM18-768, while 22.9%, 8.9% and 41.5% slower than that on AFG16-766, FM16-765 and BLS24-509. Besides, after employing degree-2 endomorphism, the whole pairing computation on GG22D7-457 is 24.6%, 10.4%, 27.0% and 22.5% faster than that on KSS16-766, AFG16-766, FM16-765 and FM18-768, while 13.1% and 49.7% slower than that on KSS18-638 and BLS24-509. The results show that curve GG22D7-457 can be regarded as an alternative choice, which depends on the protocols requirements and performance trade-off. This curve may benefit from efficient performance for the operations in $\mathbb{G}_1$ since the characteristic p of the prime field is relatively small compared to other curves at the same security level. Thus it may be an appropriate choice if the protocols require fast operations on in $\mathbb{G}_1$, such as DAA schemes [9]. Therefore, our techniques can be exploited to expand the employments for the super-optimal pairings on more pairing-friendly curves with degree- n endomorphisms, making them be more competitive in pairing-based cryptography.

5 Conclusion and Future Work

In this work, we gave detailed research for the pairing-friendly curves equipped with degree- n endomorphisms ($n > 1$), especially the curve GG22D7-457 at the 192-bit security level with several cryptographically great properties. To improve

Table 3. Benchmarking results for the pairing computations among different pairing-friendly curves at 192-bit security level, reported in 10^3 clock cycles in a 64-bit processor. The results are averaged over 10^4 executions.

Curve	Miller loop	Final exp	pairing total
GG22D7-457, $D = 7$ (sopt)	7168	15764	22933
GG22D7-457, $D = 7$ (opt)	9691	15647	25539
KSS16-766 [20], $D = 1$ (opt)	9349	21077	30427
AFG16-766 [37], $D = 1$ (opt)	5525	20082	25607
FM16-765 [16], $D = 1$ (opt)	6526	24826	31413
KSS18-638 [20], $D = 3$ (opt)	7518	12405	19922
FM18-768 [16], $D = 3$ (opt)	7813	21782	29595
BLS24-509 [20], $D = 3$ (opt)	4190	7348	11538

the performance of pairing computation on GG22D7-457, we first proposed the formulas for super-optimal pairings on the curves with degree- n endomorphisms and targeted the case $n = 2$ to this curve. Building upon this, we made a specific theoretical analysis of super-optimal pairing formulas on families GG22D7 and GG28D11, constructed by using degrees-2 and 3 endomorphisms, respectively.

Additionally, we presented efficient implementation details for pairing computation on GG22D7-457 utilizing our methods, and made a concrete computational cost analysis. Our results illustrated that using our new super-optimal pairing formula can reduce about **31.9%** $\mathbb{F}_p$ -multiplications compared to the optimal ate pairing for the Miller loop on GG22D7-457. In conclusion, our techniques can be leveraged to construct more super-optimal pairings on pairing-friendly curves with degree- n endomorphisms ($n > 1$), shortening the bit-lengths of Miller loops to $\log(r)/2\varphi(k)$. Consequently, this work extends the application of super-optimal pairings, and provide more choices for the pairing-friendly curves in pairing-based cryptographic protocols.

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